

HW9: Box-Jenkins Model Identification

ISYE 4031 - Summer 2026 - Problems Only Version

Submission Instructions

Include your full name, Georgia Tech email, and GTID in your submitted work. You may complete the homework using JupyterLite, a local Python or R environment, spreadsheet software, handwritten work where appropriate, or another approved environment. Your submission should clearly answer each question and show enough reasoning, formulas, code output, or supporting work to be graded.

Required Data Files

- shimmer_shampoo_hw9.csv

Problem Data

The following data give the number of bottles of Shimmer Shampoo sold at a grocery store over the previous 30 weeks. Use the Box-Jenkins workflow and any software of your choice to answer the questions.

Problem 1(a)

Plot the time series of the original data. Give your opinion on whether this time series is stationary.

Problem 1(b)

Find the first-order differences, plot them, and give your opinion on whether to choose the original Shimmer Shampoo sales values y_t or the first differences z_t as the stationary working series.

Problem 1(c)

Assuming that the original time series y_t is stationary, identify a tentative Box-Jenkins model for y_t . Include your decision about whether to include a constant term.

Your answer should use the ACF/PACF pattern and the mean of the working series. You may also fit the tentative model to check whether the result is sensible.

Problem 1(d)

Assuming that the original series is not stationary and the first differences z_t are stationary, identify a tentative Box-Jenkins model for z_t . Include your decision about whether to include a constant term.

Your answer should use the ACF/PACF pattern of the first differences and the mean of the first differences. You may also fit the corresponding ARIMA model on the original y_t .

AI Usage Report

Your submission must include an AI Usage Report following the course AI policy. If you used AI tools, include the tool(s), what you asked them to help with, what output you accepted, changed, or rejected, and how you verified the final work. If prompts or screenshots materially affected your submission, include copied prompt text or screenshots in the report when appropriate. If no AI tools were used, include this statement: "No AI tools were used for this submission. You may include this report at the end of your submission or attach it as a separate file.